

Retrograde Urethrogram (RUG)

Patient Information · What to expect before your X-ray of the urethra

WARWIKI

A retrograde urethrogram, or “RUG,” is a short X-ray test of your urethra. The urethra is the tube that carries urine out of your body. A small amount of dye is gently placed into the urethra so it shows up clearly on the X-ray. This helps your surgeon find and measure a narrowing, scar or injury before planning treatment.

About This Test

A scar or narrowing inside the urethra is called a **stricture**. It can be hard to see on a regular scan. A RUG fills the tube with a dye that shows up white on X-ray. This makes a clear “road map” of the inside of your urethra.

Your surgeon may order a RUG to:

- Find a narrowing or scar, and measure where it is and how long it is
- Check the urethra after an injury
- Plan a surgery to repair the urethra
- Check how the urethra is healing after a repair

The test shows the channel that urine passes through. It does not look at the kidneys, and it is not a treatment. It is a picture that guides what comes next.

Is It Safe?

Yes. A RUG is a common, low-risk test. It uses a small amount of X-ray. The dye is water-based and leaves your body in your urine. Most people have only brief, mild discomfort. A urine infection, a little bleeding, a reaction to the dye, or trouble urinating for a short time can happen, but these are uncommon.

Learn the Terms

Urethra

The tube that carries urine from the bladder out of the body.

Stricture

A narrow or scarred spot in the urethra that can slow or block urine.

Contrast dye

A liquid that shows up white on X-ray so the inside of the urethra can be seen.

Catheter

A thin, soft tube. For this test, only the tip sits just inside the opening.

Fluoroscopy

A type of X-ray that shows live, moving pictures as the dye flows.

Meatus

The opening at the tip of the penis where urine comes out.

Urine culture

A test on a urine sample to check for infection.

Reconstructive urologist

A surgeon who repairs the urinary tract, including the urethra.

Will it hurt? Most people feel only brief pressure or stinging as the dye goes in, not sharp pain. Tell the team if you are uncomfortable; they can slow down. The test is over in about 15–30 minutes.

How to Get Ready

There is very little to do. In most cases:

- You can **eat and drink normally**, unless your team tells you otherwise.
- Take your **usual medicines** unless your team asks you to stop one.
- You can usually **drive yourself home**. Most people do not need medicine to relax.

Tell your team ahead of time if you:

- Have had a reaction to X-ray dye
- Take a blood thinner
- Stretch the urethra yourself, have a catheter, or had a recent stretch or cut. Before planning a repair, the test may be done after the urethra rests for some weeks.
- Have signs of a urine infection, such as burning or fever. The test may be moved to another day.

You may be asked for a **urine sample** to check for infection. Some teams give an **antibiotic** before or after the test.

What Happens During the Test

- 1 You lie on an X-ray table, usually turned slightly onto your side.
- 2 The tip of the penis is cleaned. A small, soft tube is placed **just inside the opening**. A tiny balloon, or a clamp on the tip of the penis, may hold it in place.
- 3 The dye is slowly placed into the urethra while X-ray pictures are taken. Hold still and breathe normally.
- 4 The team takes pictures from a few angles. Then the tube is removed.

After the Test

- You can usually go back to **normal activities right away**.
- You may have **mild burning when you urinate** for a day or so. Drinking extra water helps.
- A **small amount of blood** at the tip, or pink urine, is common. It usually clears within a day.

Call the office that did your test if you have:

- A temperature of **100.4 °F (38 °C)** or higher, or chills
- Burning or pain that gets worse instead of better
- Bleeding that lasts more than a day

Go to an emergency department if:

- You cannot urinate and your bladder feels full or painful
- Heavy bleeding, or urine like red wine or with clots
- You cannot reach the office and are worried

Three Things to Remember

1. A RUG is a quick X-ray that maps the inside of your urethra.
2. Tell your team about any dye reaction, blood thinner or urine infection.
3. Mild burning or a little blood for a day is normal. Drink extra water.